DATA 102

Modeling Degree ROI Through University Financial Data

MCO PHASE 1

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PHASE 1

- Libraries & Data Set
- Data Cleaning
- Exploratory Data Analysis (EDA)
- Research Question
- Statistical Inference

Data Set & Libraries

ABOUT THE DATA SET

College Scorecard Subset

- *Source: U.S Department of Education*
- *Observations: 6,273 institutions*
- *Variables: 24 institutional and financial attributes*

KEY VARIABLES

- *In-state tuition and fees*
- *Median earnings 10 years after entry*
- *Institutional control type (public, private nonprofit, for-profit)*
- *Median graduate debt*
- *Undergraduate enrollment*

LIBRARIES

Pandas & Numpy

- *Data cleaning, manipulation, and numerical operations*

Matplotlib & Seaborn

- *Data visualization and exploratory analysis*

Scipy.stats

- *Statistical testing (normality tests, correlation analysis)*

Statsmodels

- *Statistical modeling and diagnostic tools*

Data Cleaning

- SAT scores (83.5% missing) and admission rates (69.6% missing) were excluded from analysis due to high levels of data gaps
- Converted “PrivacySuppressed” values in graduate debt to missing (NaN)
- Dropped observations missing tuition, earnings, or graduate debt
- Removed institutions with $0 in-state tuition
- Imputed missing undergraduate enrollment using sector-specific medians
- Removed duplicate institutions (UNITID)
- Applied IQR-based outlier removal:
  - Earnings (global)
  - Tuition (within sector)
- Created CONTROL_LABEL for interpretability

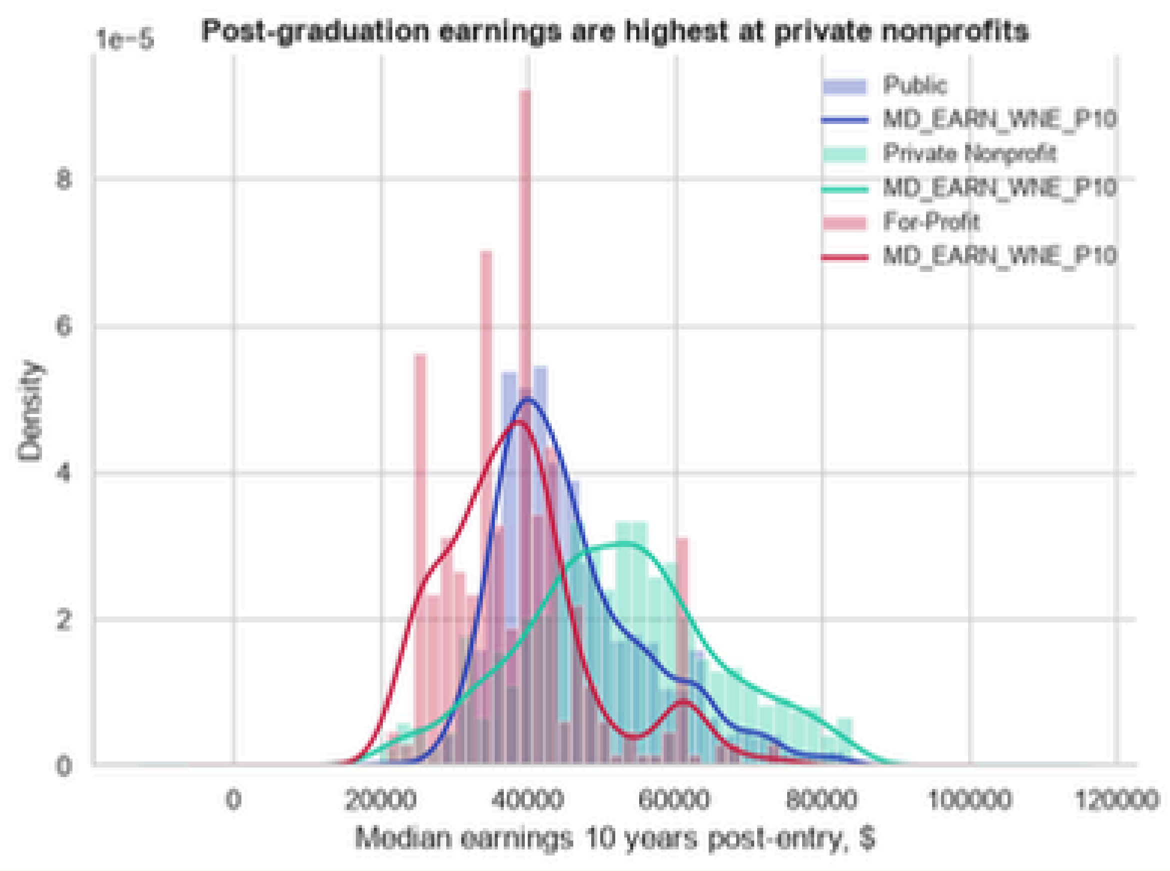
FINAL DATA SET: 2,894 *institutions*

Exploratory Data Analysis (EDA)

Exploring six (6) questions examining:

- Earning and tuition distributions
- Cross-sector comparisons
- Tuition-earnings relationship
- Debt and selectivity effects
- Equity differences across Minority Serving Institutions (MSIs)

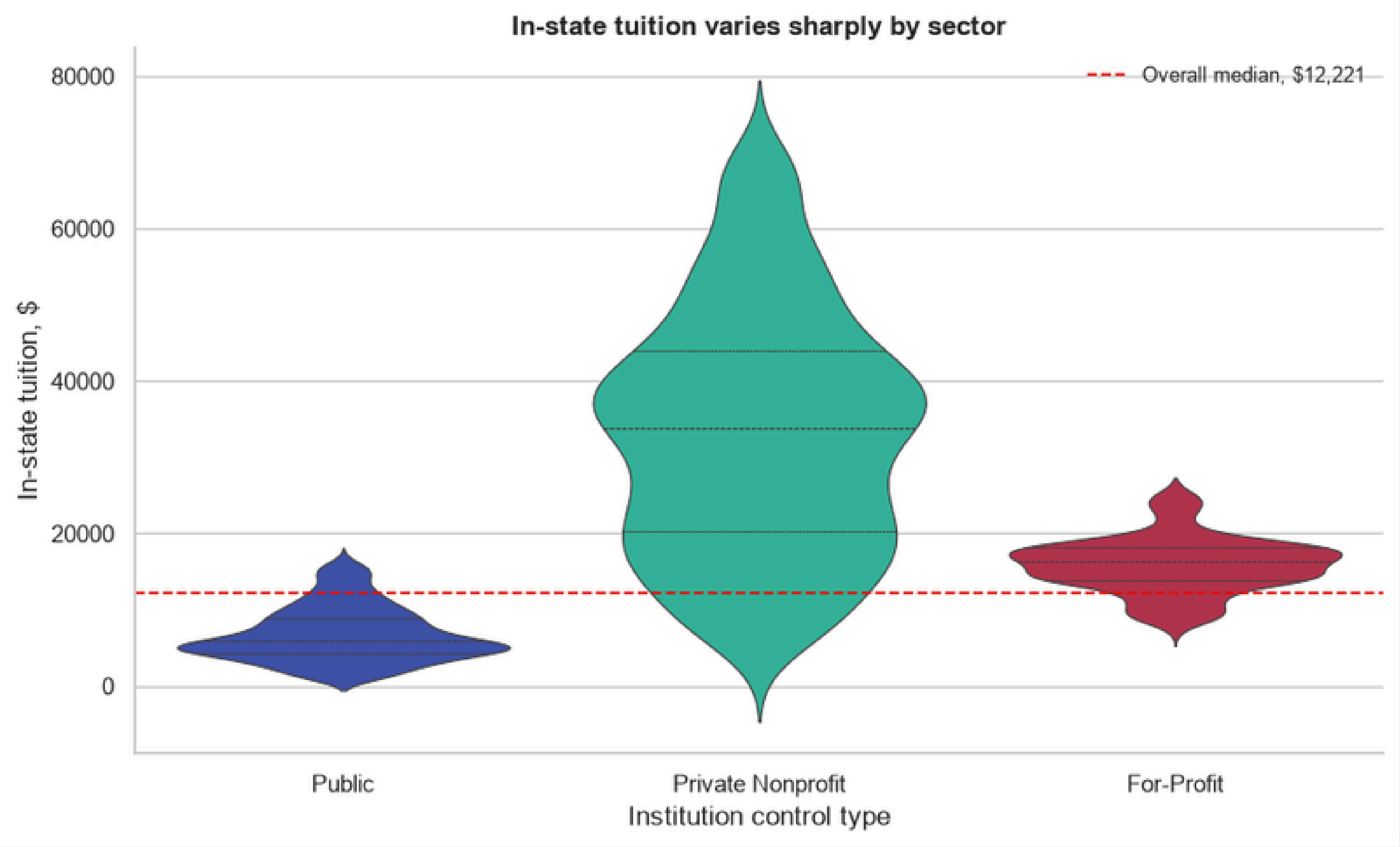
Exploratory Data Analysis (EDA)



Earnings Distribution By Sector

- Private nonprofit institutions had the highest median and mean earnings
- For-profit institutions had the lowest median earnings
- Private nonprofit institutions showed the greatest variation in earnings
- Earnings distributions overlapped across all three sectors

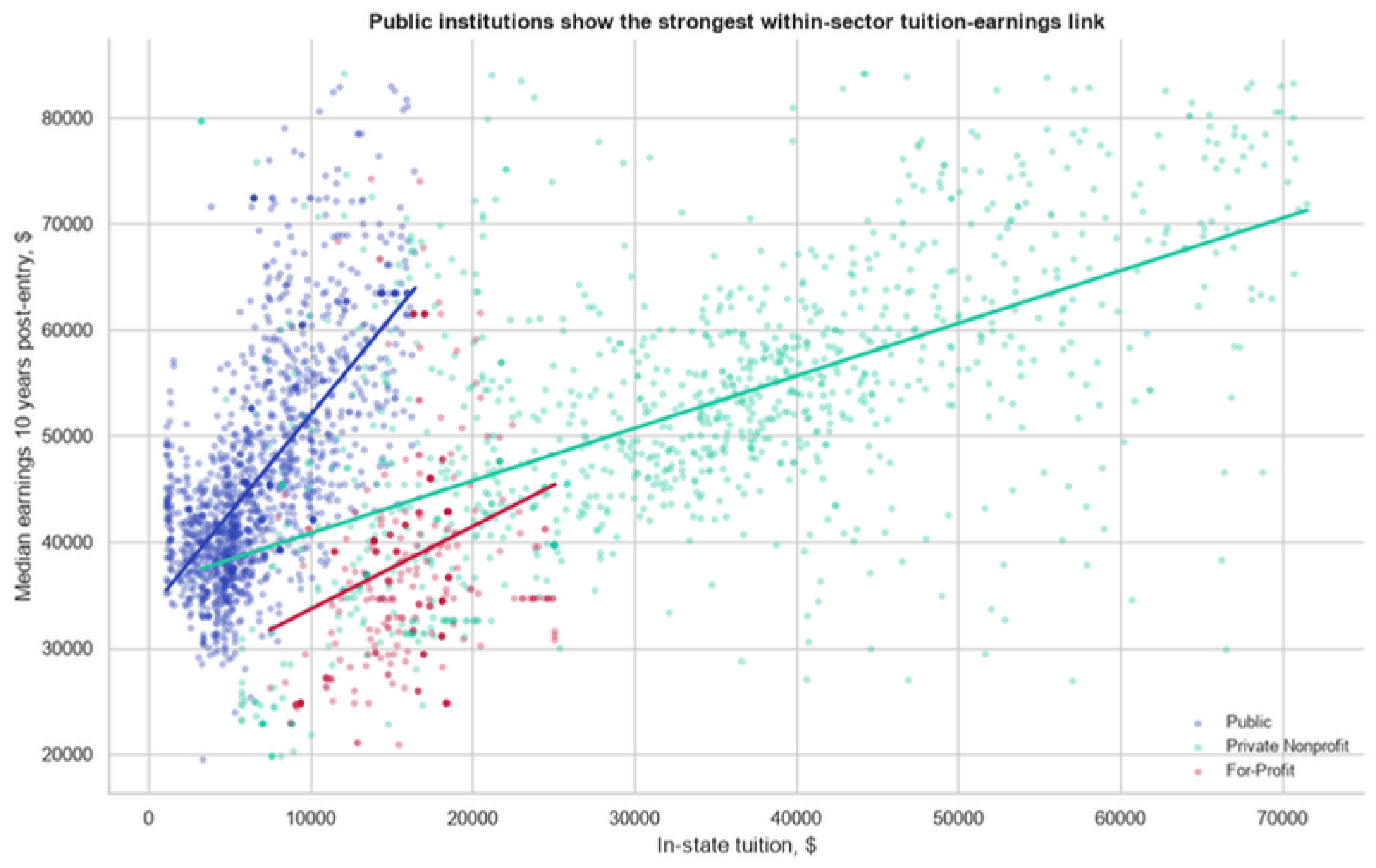
Exploratory Data Analysis (EDA)



Tuition Distribution By Sector

- Private nonprofit institutions charged the highest tuition
- Public institutions had the lowest tuition
- For-profit tuition was between public and private nonprofit institutions
- Tuition distributions were more separated than earnings distributions

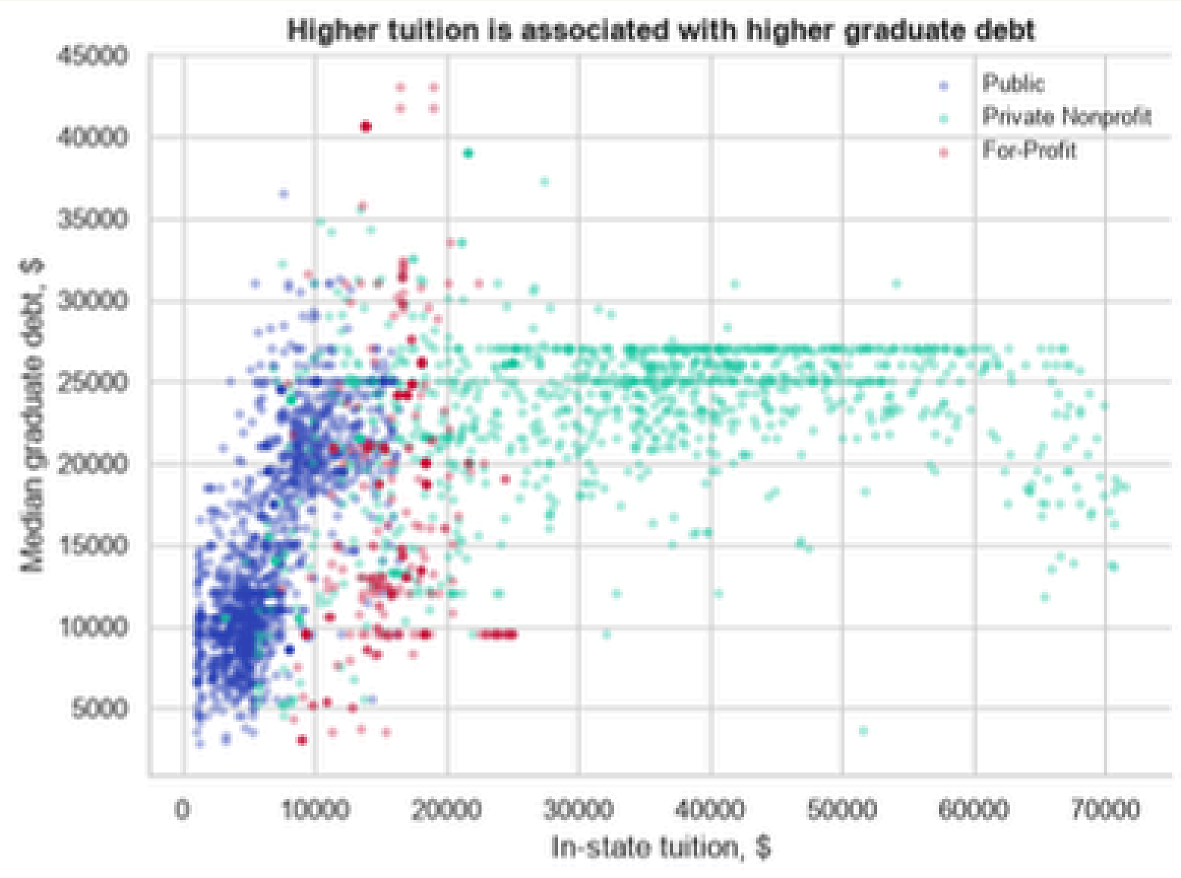
Exploratory Data Analysis (EDA)

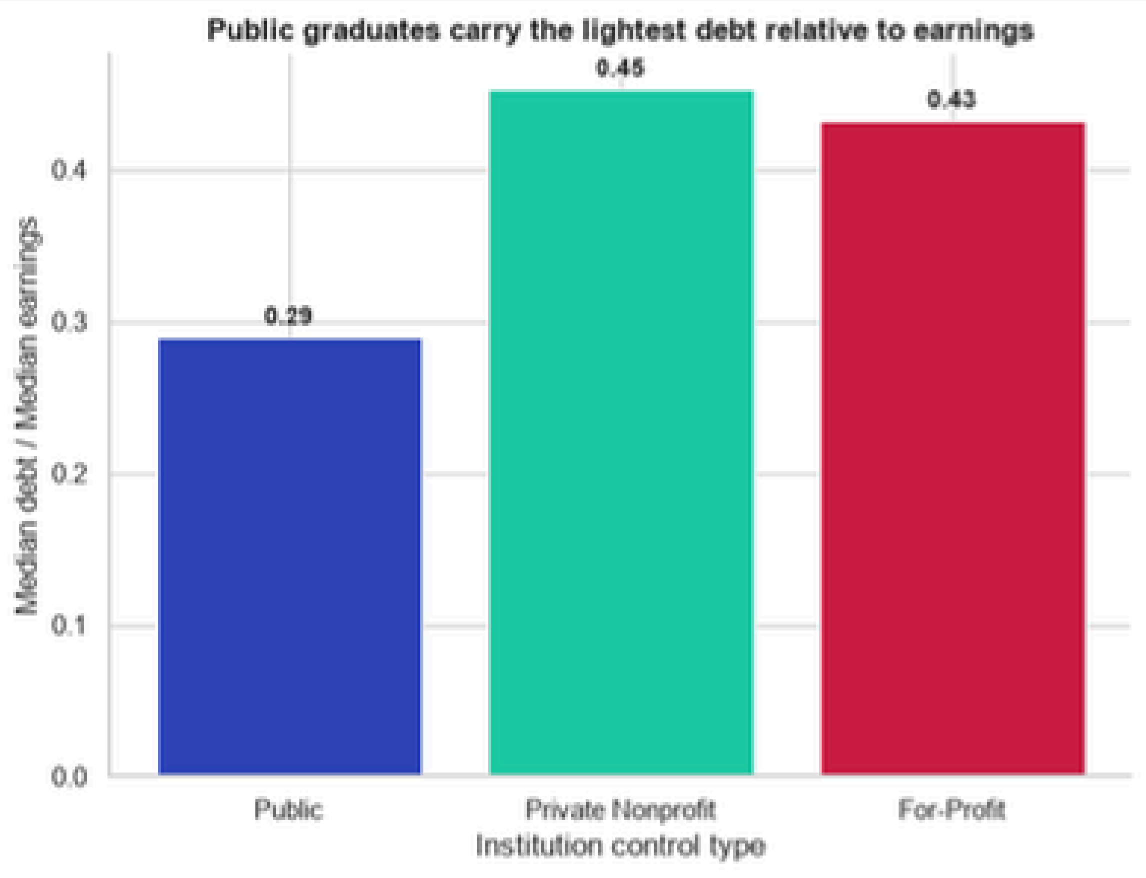


Joint Tuition-Earnings Relationship

- Tuition and earnings were moderately positively correlated
- Public institutions showed the strongest within sector correlation
- For-profit institutions showed the weakest correlation
- Institutions formed distinct clusters by sector

Exploratory Data Analysis (EDA)





Debt As A Mediating Variable

- Higher tuition was associated with higher graduate debt
- Public institutions had the lowest median debt to earnings ratio
- Private nonprofit and for profit institutions had higher debt burdens relative to earnings

Exploratory Data Analysis (EDA)

Admissions Selectivity and Earnings by Sector

- Higher admissions selectivity is consistently associated with higher graduate earnings across all sectors.
- The positive correlation between tuition and earnings persists even when comparing schools of similar selectivity.
- Selectivity explains some of the variance, but it does not fully account for the tuition-earnings relationship in the broader market.

Minority-Serving Institution Profiles by Sector

- Minority serving institutions generally charged lower tuition within each sector
- Minority serving institutions generally reported lower median earnings
- Most minority serving institutions were in the public and private nonprofit sectors

Note: The For-Profit sector was severely underrepresented in admissions reporting data.

Research Question

Is there a significant relationship between in-state tuition and median post-graduation earnings, and does this relationship differ across public, private nonprofit, and for-profit institutions?

Statistical Inference

To test the relationship between in-state tuition and post-graduation earnings, we conducted a correlation analysis with an assumption check for normality.

Hypotheses

- H_0: There is no linear relationship between in-state tuition and median post-graduation earnings ($r = 0$)
- H_a: There is a significant relationship between in-state tuition and median post-graduation earnings ($r \neq 0$)
- Significance level: $\alpha = 0.05$

Methodology

- Normality was tested using the D'Agostino K-squared test
- Since earnings data were not normally distributed ($p < 0.05$), we used Spearman rank correlation instead of Pearson correlation. It does not assume normal distribution, captures monotonic relationships, more appropriate for skewed real-world data.

Results

- Normality test p-value: 0.0000
- Spearman correlation coefficient (r): 0.4641
- p-value: 1.53×10^{-154}

Conclusion

- Reject the null hypothesis. There is a statistically significant moderate positive relationship between in-state tuition and post-graduation earnings.

DATA102

Thank You!

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